# Practice workbook · Calculate area, volume and surface area to volume ratio

Cube side lengths or a measured circle are given; surface area, volume, their ratio, or a circular area is required.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-AREA-VOLUME-SA-V

# 1. Question 3 · part a.i · 3 marks

January 2023 Paper 2

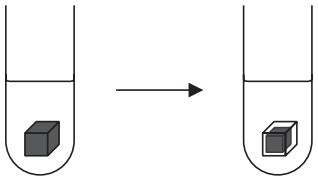
Worked-example retry · Attempt part a.i only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

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## 1. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

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## 2. Question 6 · part c.i · 3 marks

June 2019 Paper 1

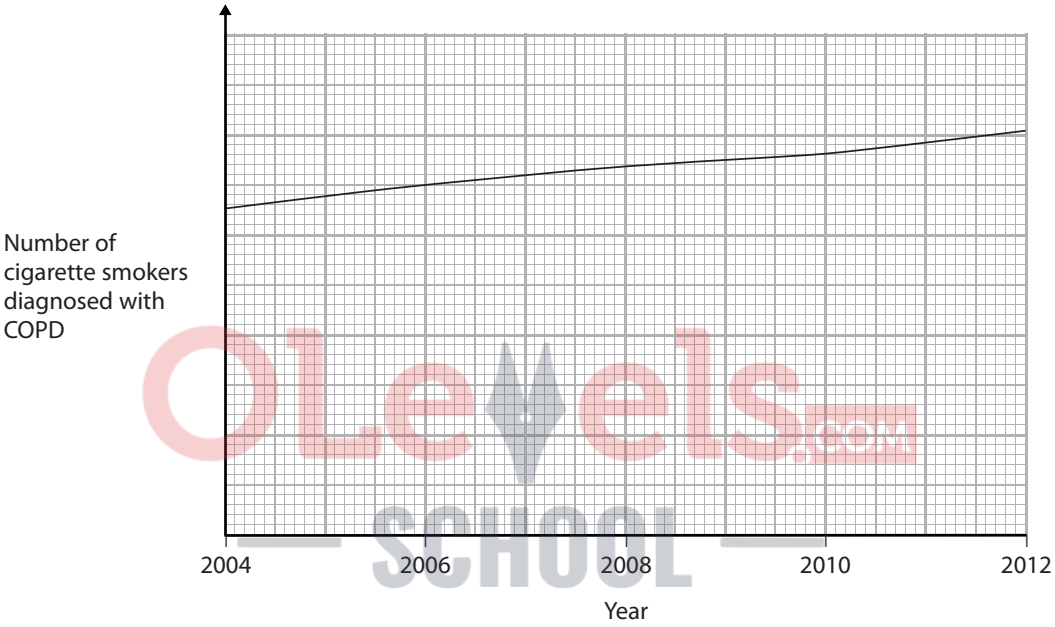
Further practice · Attempt part c.i only.

**6** (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.




(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

**(1)**

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

**(2)**

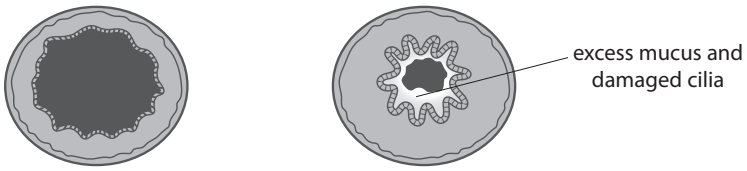
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## 2. Question 6 · part c.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part c.i only.

(b) The diagram shows how chronic bronchitis affects the airways in the breathing system.




healthy airway

chronic bronchitis

Explain how excess mucus and damaged cilia affect the breathing system of a person with chronic bronchitis.

**(2)**

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## 2. Question 6 · part c.i · 3 marks

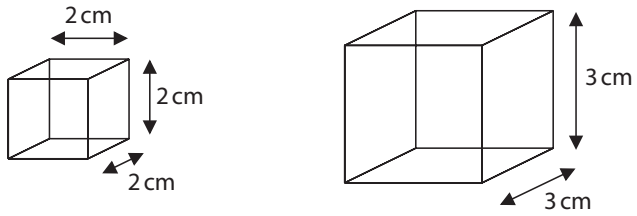
June 2019 Paper 1

Further practice · Attempt part c.i only.

(c) The diagram shows models of two alveoli.

One model represents an alveolus from a person with healthy lungs.

The other model represents an alveolus from a person with emphysema.




**Healthy** **Emphysema**

The table shows the surface area to volume ratio for a healthy alveolus.

| | Surface area in $cm^2$ | Volume in $cm^3$ | Surface area to volume ratio |
|---|---|---|---|
| Healthy | 24 | 8 | 3 : 1 |
| Emphysema | | | |

(i) Complete the table by giving the missing information.

**(3)**

(ii) Explain how the surface area to volume ratio of alveoli in the lungs of a person with emphysema will affect the normal function of body cells.

**(3)**

**(Total for Question 6 = 11 marks)**

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## 2. Question 6 · part c.i · 3 marks · Working and answer

Working and answer

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## 3. Question 6 · part a.i · 2 marks

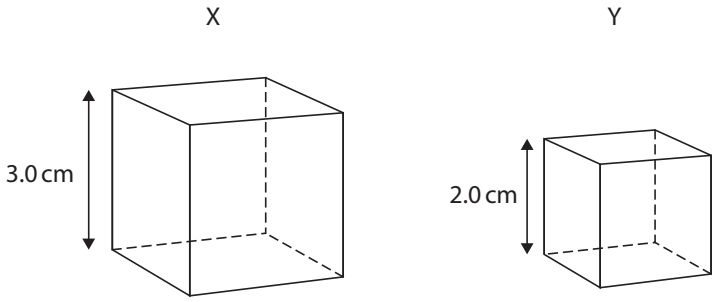
June 2024 Paper 1

Further practice · Attempt part a.i only.

**6** (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.




The table gives the SA : V of cube Y.

| Cube | Surface area in $cm^2$ | Volume in $cm^3$ | SA : V |
|---|---|---|---|
| X | | | |
| Y | 24.0 | 8.0 | 3:1 |

(i) Calculate the surface area of cube X.

**(2)**

surface area = ..... $cm^2$

(ii) Calculate the volume of cube X.

**(2)**

volume = ..... $cm^3$

(iii) Calculate the SA : V of cube X.

**(2)**

SA : V = ..... : 1

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## 3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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## 4. Question 6 · part a.ii · 2 marks

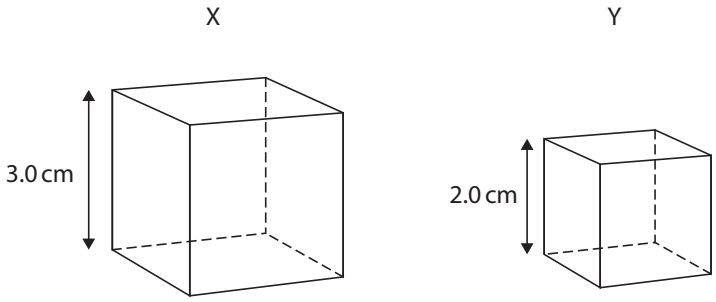
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Further practice · Attempt part a.ii only.

**6** (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.




The table gives the SA : V of cube Y.

| Cube | Surface area in $cm^2$ | Volume in $cm^3$ | SA : V |
|---|---|---|---|
| X | | | |
| Y | 24.0 | 8.0 | 3:1 |

(i) Calculate the surface area of cube X.

**(2)**

surface area = ..... $cm^2$

(ii) Calculate the volume of cube X.

**(2)**

volume = ..... $cm^3$

(iii) Calculate the SA : V of cube X.

**(2)**

SA : V = ..... : 1

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## 4. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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## 5. Question 6 · part a.iii · 2 marks

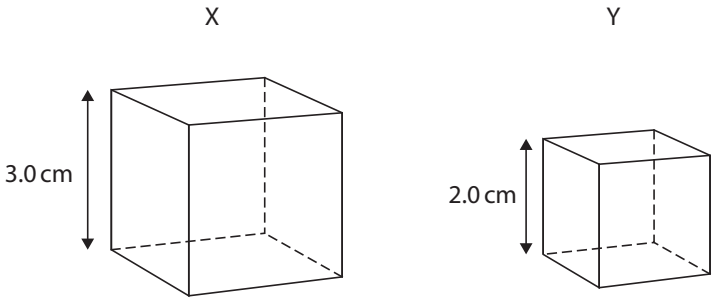
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Further practice · Attempt part a.iii only.

**6** (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.




The table gives the SA : V of cube Y.

| Cube | Surface area in $cm^2$ | Volume in $cm^3$ | SA : V |
|---|---|---|---|
| X | | | |
| Y | 24.0 | 8.0 | 3:1 |

(i) Calculate the surface area of cube X.

**(2)**

surface area = ..... $cm^2$

(ii) Calculate the volume of cube X.

**(2)**

volume = ..... $cm^3$

(iii) Calculate the SA : V of cube X.

**(2)**

SA : V = ..... : 1

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# 5. Question 6 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

**(5)**

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## 5. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

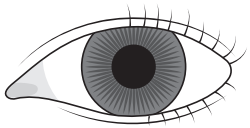
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# 6. Question 6 · part b · 3 marks

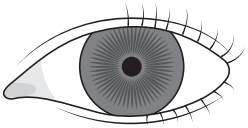
June 2024 Paper 2

Further practice · Attempt part b only.

**6** The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

**(3)**

(b) A student calculated the area of the pupil in bright light to be 7.1 $mm^2$.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = $\pi r^2$ $\pi = 3.14$]

**(3)**

area of pupil in dim light = ..... $mm^2$

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## 6. Question 6 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

**(3)**

(ii) Explain the adjustment made to the pupil when moving into dim light.

**(2)**

**(Total for Question 6 = 11 marks)**

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## 6. Question 6 · part b · 3 marks · Working and answer

Working and answer

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